Testing What a Sentence Expander Must Preserve: Input-Conditioned Requirements and a Tested Oracle for LLM-Based AAC

[PENDING: author names]
[PENDING: affiliations, emails]

***Abstract*—Given the complete typed sentence "Wear headphones for your music.", an LLM sentence expander for augmentative and alternative communication (AAC) returned "Please wear headphones for your music." Across four models and 10 authored directive families, fully typed plain sentences gained a politeness marker in 9/40 cases, while polite ones lost theirs in 2/40; per model, additions ranged from 0/10 to 6/10, and an explicit preservation prompt changed the balance (4/40 vs. 6/40). Such failures are hard to test because expansion has no single correct output, and keywords often leave meaning open. We write *input-conditioned* preservation requirements: they score a commitment such as negation, a question, or plainness only when the input supplied it, and report the model's choice as a default when it did not. Executable checks over controlled input channels, plus a partial oracle with blind extraction and code-computed verdicts, separate interface loss from model violation. We also test the oracle: an identity check exposed a specification defect, and the obvious relaxation then missed keyed statement-substitution mutants, so the affected families are quarantined. A developer can act on these checks: exact-copy routing passes all 116 non-quarantined typed sources at zero model calls, and a runtime guard repairs 63/80 supplied-cue failures, leaving 17. All rates are machine-scored on constructed items; validation against human labels is in progress.**

***Index Terms*—accessibility, AAC, LLM testing, requirements, test oracle, metamorphic testing**

I. Introduction

People who use AAC often compose a message by selecting a few words or symbols, because every selection costs time and effort [1], [2]. LLM keyword expansion turns such a selection into a full sentence [3], [2]. The same feature is sometimes given a sentence the person has already typed in full. Fig. 1 shows what four models did then, on 10 directive families written in plain and polite versions. In 9/40 plain cases the output added a marker such as "please"; the reverse happened in 2/40 (paired difference +17.5 points [+5.0, +32.5]). The behavior is concentrated, not universal: Qwen added 0/10, Llama 1/10, Gemma 6/10, and gpt-6.1-terra 2/10. It is also prompt-dependent: under a cached prompt that explicitly forbids added politeness, additions fell to 4/40 but removals rose to 6/40.

Adding "please" need not be an error in general; politeness suggestions can change how writers phrase messages and how they perceive a relationship [4], [5]. But here the person had already supplied the wording, so a system that rewrites it is speaking for them. Contrast `like medication`: these keywords do not say whether the person likes the medication,

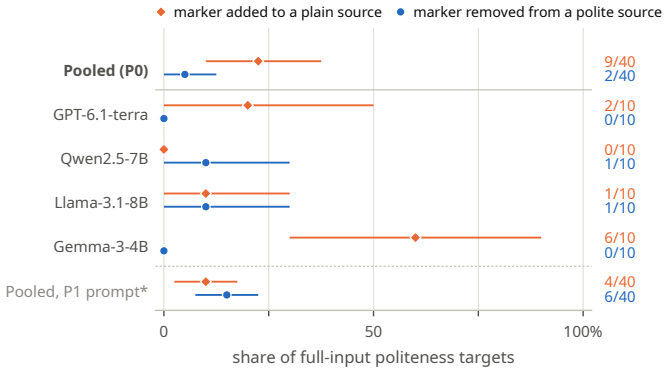


Fig. 1. Full input (the source sentence is visible to the model). Politeness markers added to plain sources versus removed from polite ones, P0 prompt, ten politeness families, 95% family-bootstrap intervals; counts at right. The P1 row is a cached sensitivity under a preservation instruction (not a primary arm). Zero observed events do not establish zero risk.

and every model chose the affirmative. No model can be blamed for that guess, but the interface decides who has to pay an extra selection to say "don't". Testing such a feature therefore needs requirements that depend on what the input actually carried.

We treat meaning preservation in AAC expansion as an accessibility requirement that should be specified, tested and regression-checked. We make three contributions. (1) *Input-conditioned requirements* (§III): executable checks that score a commitment only when the visible input supplied it, run over controlled input channels of minimal-pair families. (2) *A partial oracle that is itself tested* (§IV), with a stated failure envelope, including a defect our own repair introduced. (3) *A worked requirement → check → developer decision → failure envelope sequence* (§VII), including an evaluated runtime guard and an exact-copy baseline. We ran no study with AAC users and make no claims about benefit, effort or preference. All rates are machine-scored and remain provisional until the human validation in §IV is complete.

II. Related Work

Testing LLM systems: CheckList organizes behavioral tests into minimum-functionality, invariance and directional tests [6]; Cho et al. catalogue metamorphic relations for

LLM-based NLP [7]; COLLIE tests compositional generation constraints [8]. Sensei combines a conversational user simulator with a DSL for assertions and metamorphic relations, evaluated on seeded faults and deployed chatbots [9]. Extracting fields, asserting properties and relating test cases are therefore established; we do not claim a new testing technique. Our contribution is the AAC input specification: identical keywords for alternative readings, controlled cue availability, and separate treatment of lost information, violated supplied cues and defaults. LLM judges show position, verbosity and self-preference biases [10], [11], and similarity metrics can miss negation [12]; we restrict the judge to blind extraction and test it.

LLMs in AAC: Expansion and suggestion tools include KWickChat [3], SpeakFaster [2] and systems studied with AAC users [1]. Several expose intent controls: intent buttons [13], intention choices for people with aphasia [14], stance tags [15]. Lacosse et al. keep bracketed function words such as "not" and evaluate semantic equivalence [16]; Choudhury et al. test whether affect cues reach the output [17]; Gorenshtein et al. benchmark intent drift, including negation, under simulated decoder errors [18]. R-SPEAK was co-designed with people with aphasia and scores relevancy, faithfulness and completeness on real aphasic speech [19]. So AAC generation already has controls and fidelity evaluation; what we add is a conditional specification of *which* commitments are testable under each input, and a test of the oracle. Outside AAC, Navneet et al. show LLM rewriting can keep semantic similarity while erasing pragmatic markers, and that preservation prompts reduce but do not remove this [20]; our P1 sensitivity has the same shape. Accessibility engineering spans the life cycle, not only interface audits [21]; conformance checks cannot see whether generated output keeps the user's meaning.

III. Input-Conditioned Requirements

Families and channels: A *family* is a minimal pair of authored source messages that share content words and differ on one *critical* dimension: polarity, agent (who acts), certainty, speech act, politeness, or time/quantity. For example, "I don't like the new medication" and "I like the new medication" both compress to `like medication`. 2 speech-act families (refusal/acceptance) also differ in polarity; they are quarantined from primary results (§IV). Each variant is rendered through five channels (Fig. 2): *bare* keywords; *noncrit*, which adds one family-specific content word (`like new medication`); *chip1*, which adds the critical cue as a bracketed tag (`[NOT]`); *chipall*, with every tag the variant needs; and *full*, the source itself. Every variant needs at least one tag, so the unmarked reading gets a counter-tag (e.g., `[PLAIN]`) and neither variant passes for free. Tags are a test device for supplying a cue, not a proposed interface.

Requirements: For output y from input x for variant v, let $L_d(y)$ be the extracted value on dimension d and $C(x)$ the commitments x carries. **R1** (supplied cue): if $C(x)$ fixes d to c, then $L_d(y)$ must be in the admissible set for c; otherwise it is a *Violation*. **R2** (full input): if x is the full source, each scored explicit value of the source is kept, including the *absence* of politeness markers. **R3** (ambiguity): if $C(x)$ does not fix d, $L_d(y)$ is not pass/fail; a committed value is logged as an *Assumption*, and its direction is reported, because a systematic default decides which reading is free. **R4** (composition): adding a compatible cue for e should not change a satisfied value on $d \neq e$. R1, R2 and R4 concern commitments the input supplied. R2's plainness clause is a software policy for typed text, not a claim about the author's private intent: absence of "please" may or may not be deliberate, and we only require that the system not change it unasked. *Loss* names the case in which a channel never carried the value.

Executable checks: MR2 (cue restoration) and MR3 (full preservation) are target-based contract checks on chip1/chipall and full outputs: they assert R1/R2 on a single output. MR4 relates a single-cue output to a two-cue output and counts pass→fail transitions (R4). MR1 compares bare with noncrit. Because both inputs leave d open, a changed label is a *default-sensitivity diagnostic*, not a defect; it also serves as a one-selection control when reading the cue effect. Treating it as an invariance requirement would need extra applicability assumptions that some words violate ("weekend" and "tonight" carry time). MR0 tests the oracle: an unchanged source must pass.

What the conditioning changes: A reference-fidelity score compares each output with its authored source. It would fail "I like medication" against "I don't like the new medication" although the keywords never carried the negation; our checks record it as an Assumption and charge the interface, not the model (E1). Conversely, a similarity score would rate "Please wear headphones for your music." as close to the typed sentence; R2 fails it because the input supplied the plain wording (E3). The same output can thus be a default under one channel and a violation under another, which is the distinction a developer needs before deciding what to fix.

With a balanced pair, any bare-channel output matches at most one of the two variants, so 50% is a design ceiling for paired match, not a target or a prior over real intentions.

IV. A Partial Oracle and Its Failure Envelope

The oracle decides only whether a fixed set of explicit, text-level values is kept. It does not judge fluency, appropriateness, referents or whether anyone would accept the sentence.

Pipeline as run: An extractor (gpt-6.1-sol, a different model from the GPT model under test) sees one output and answers the family's critical question plus five general questions; it never sees source, model, prompt or channel. Two prompt variants (J1, J2) differ in wording, option order and item order. Critical labels are resolved once per unit (text, visible input, dimension) and shared across runs: a cross-family arbiter (claude-opus-5-5) decides J1/J2 disagreements, otherwise the agreed label wins, otherwise the label is unknown; there is no single-prompt fallback. The arbiter sees the output, the codebook and the visible input, so for arbitrated full-channel units it is not blind to the source; this is a disclosed exception

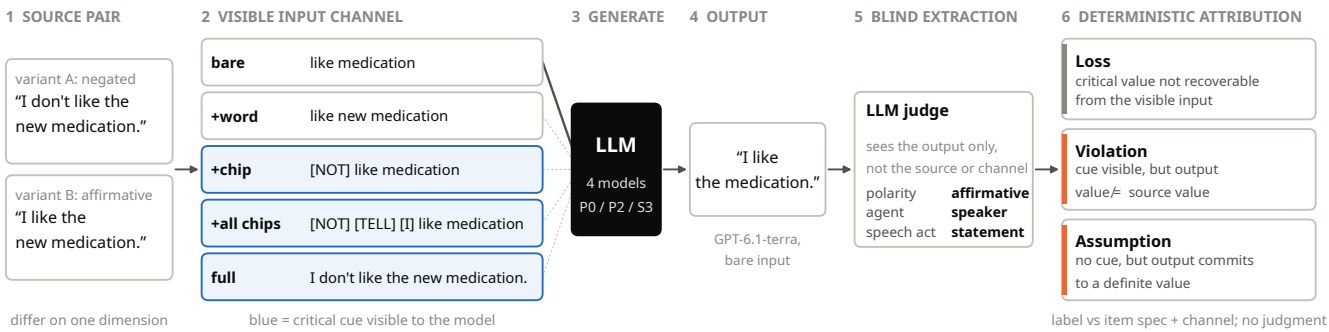


Fig. 2. Harness, traced for one stored output. Family pol-07 has two source variants that share the keywords. Each variant is rendered through five channels; the shown output (gpt-6.1-terra, P0) came from the *bare* channel only. In *full*, the source is visible to the generator; extraction is always blind to source, model and channel. Verdicts are computed in code from labels, item specification and channel; Loss, Violation and Assumption have separate denominators.

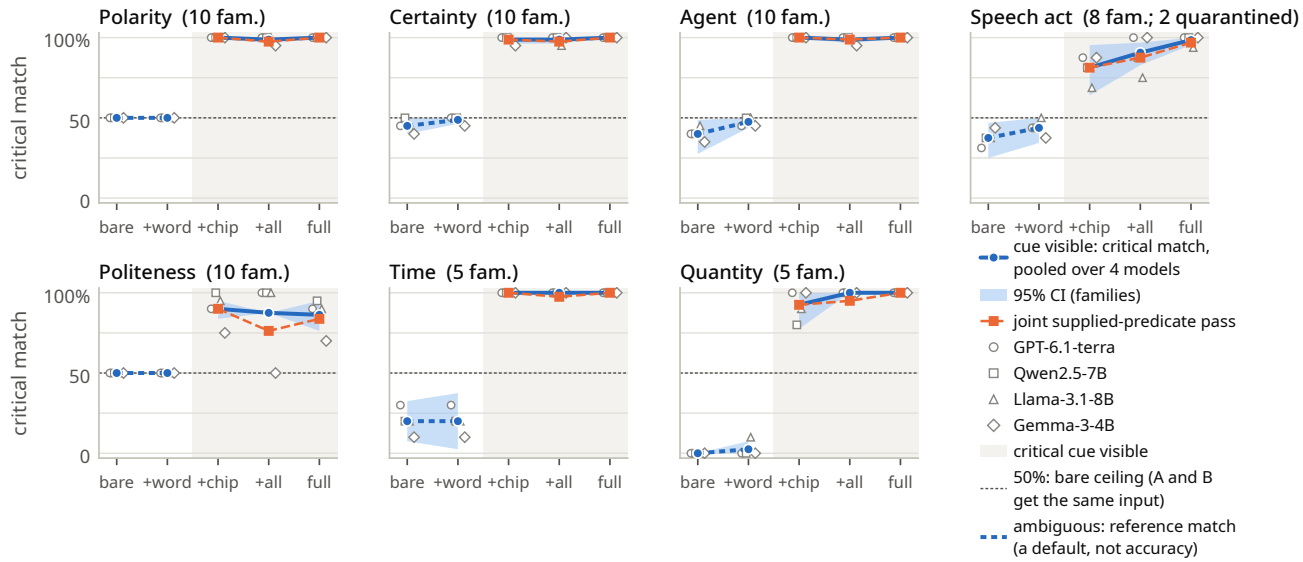


Fig. 3. Critical-value match by channel (P0, four models, 58 primary families; speech act shows the 8 non-quarantined families). Dotted segments (bare, +word) are ambiguous: their match is a chosen default bounded by the 50% paired ceiling, not accuracy. Solid segments assess supplied cues; the dashed orange line adds joint preservation of all scored supplied predicates. Markers: individual models.

to blind extraction. In the main run 155 scored rows (80 units) were arbitrated and 0 remain unknown. General-dimension disagreements are broken by a rule-based extractor (layer0), which the guard in §VII also uses; this shared component limits their independence. Verdicts are computed by code from labels, the item specification and the channel, so a rule change re-scores all outputs without model calls.

Testing the oracle: Table I lists the evidence. Each of the 120 authored test sources was injected as its own full-channel output (identity) and as one hand-written, single-value mutation, frozen before judging. A second, hand-written challenge set has four groups of six controls and six one-feature mutants with keys hashed before the first judge call. The pre-specified gate ($\geq$22/24 mutants detected, $\leq$1/24 false alarms) passes when pooled, with no slack. The protocol's wording "22/24 per new group" is impossible with six mutants per group; we report the pooled reading as an interpretation and show the group that fails it. Machine agreement is reported separately and is not an error bound: three machine coders and the judge share two errors on round-1 challenges ("I can't wait…" coded negated; "a couple of" coded as two).

TABLE I
ORACLE EVIDENCE STATUS. AUTHORED INPUTS, WILSON 95% INTERVALS; THESE ARE MINIMUM-FUNCTIONALITY CHECKS ON CLEAN SENTENCES, NOT ERROR RATES ON MODEL OUTPUTS.

Check	Result	95% CI
Identity: unchanged source, contract false alarms	0/120	[0.0, 3.1]%
Scoped single-value mutation detected	120/120	[96.9, 100.0]%
New keyed mutants detected (pooled gate)	22/24	[74.2, 97.7]%
New keyed controls, false alarms	0/24	[0.0, 13.8]%
acceptance group: mutants detected	4/6	[30.0, 90.3]%
same, without the acceptance relaxation	24/24	(all groups)
Machine panel: Violation flips (units)	2/99	not a bound
Human gold, adjudicated ($\kappa \geq 0.6$ gate)	see text	

A defect found, a repair, and its blind spot: Before these injections, the identity relation run on stored model outputs found that the original oracle failed 14 of 2,292 case- and punctuation-normalized reproductions of the source, all on one sentence, "Yes, you can borrow my car this weekend", coded *statement* against the target *acceptance* (E10 in Table II). The categories overlap, so the contract demanded a distinction the text need not force. We relaxed the rule so that *statement* satisfies an acceptance target. The keyed challenges then showed the cost: the relaxed oracle detects 4/6 acceptance mutants and misses both statement substitutions ("You borrowed my car this weekend."), whereas the raw rule detects 24/24 new mutants. Fixing a false alarm removed fault sensitivity. We therefore quarantine both refusal/acceptance families from every primary result (leaving 58 of 60 families) and report them only as sensitivity. A symmetric acceptance/refusal predicate remains open.

Human validation: [PENDING: Insert human results: per-dimension κ with CI, positive/negative counts, verdict confusion, natural vs. challenge items separately; apply the $\kappa \geq 0.6$ gate per dimension and narrow every claim for a failing dimension.]

V. Audit Setup

The test set has 60 families (10 per dimension; time and quantity five each), authored to a fixed specification and verified by an independent reviewer; prompts and codebook were developed on 24 pilot and 12 development families. The items are constructed, not messages written by AAC users. Three open models run locally with 4-bit quantization (Qwen2.5-7B-Instruct, Llama-3.1-8B-Instruct, Gemma-3-4B-it) and one API model (gpt-6.1-terra). All prompts describe the sentence-composition feature of an AAC app and include the same tag legend. P0 asks for one sentence (greedy); P2 asks for three suggestions that cover different meanings; S3 draws three samples from P0 at temperature 0.7. P1, a preservation instruction, was dropped after the pilot (its effect was at our kill line); its cached test outputs are reported only as sensitivity. The primary pool has 16,238 scored rows from 9,280 requests (P0/P2/S3); rows are not independent. Intervals are 95% percentile intervals from 2,000 whole-family bootstrap resamples (seed 20261031); A/B-shared bare requests count once for defaults. Every displayed example is a stored output selected by a declared rule, shown with the rate of its eligible stratum.

VI. Results

A. Full-input preservation (R2)

Fig. 1 gives the result. Across all primary families whose source has no politeness marker, 10/424 full-input outputs added one, so the effect is concentrated in directives. Hedges, a parallel stance marker, were neither added (0/40) nor removed (0/40) in the certainty families. Under the joint check of all scored supplied predicates, full input passed 449/464; exact copying passes by construction (§VII).

B. Channel attribution (R1, R3)

Fig. 3 separates the two kinds of evidence. Under bare input, every greedy polarity output was affirmative (40/40); certainty outputs were unhedged in 36/36 committed cases, and the speaker was the agent in 27/40. These are Assumptions (R3), not failures. Asking for three different meanings (P2) produced both polarity readings in only 10/40 suggestion sets. Adding the noncritical word left the critical label unchanged in 187/232 aligned outputs (160/192 without the time/quantity families, whose added words can carry time); as a default-sensitivity diagnostic this shows that defaults are unstable, not that the model erred.

Once supplied, cues mostly worked. Critical match rose from 194/464 (noncrit) to 440/464 with one tag, 445/464 with all tags and 452/464 with full input; all 40 negated targets were negated with `[NOT]`. The focal check overstates preservation when several cues are present: with all tags, joint preservation of every scored supplied predicate was 429/464 against 445/464 for the focal value. Speech act was the weakest cue: `[?]` kept a question in 22/28 cases (E4). Pricing the default as one selection per tag (P0) gives 0.20 [0.13, 0.27] nominal extra selections for the default reading against 0.93 [0.88, 0.96] for the other (50 families with an identified default). These are nominal: 21/284 bare misses still missed after the tag, and reading or navigation time is not counted.

C. Composition (R4)

In a follow-up on 8 new factorial scenarios (two greedy draws), adding a polarity tag never broke a satisfied polarity value (0 pass→fail of 252), but adding it to an actor tag broke 29 of 244 satisfied agent values (Qwen 19, Llama 6, Gemma 4, gpt-6.1-terra 0) while repairing 12; net change -6.6 points [-12.5, -1.6], clustered by scenario. Many failures literalize the constant `[TELL]` tag ("I don't tell you to…"), and their scoring depends on a codebook rule for embedded actors, so we read this as an encoding hazard. In a second follow-up reusing four politeness test sources, adding `[ASK]` to `[PLAIN]` lowered plain preservation from 56/64 to 41/64 (two prompts, two draws), entirely in Gemma and Llama. Greedy re-runs reproduced identical text in only 44.4–71.4% (API) and 76.4–90.8% (local) of requests, so a matched repeat-disagreement baseline is needed. [PENDING: W-D (`[TELL]` vs. `[STATEMENT]`, protected-cue pass→fail vs. same-input repeat disagreement) not yet reported; until it is, we make no general interference claim.]

D. Beyond the discovery items

[PENDING: W-C (240 fresh frozen families) not yet reported.] It also tests, as a pre-specified contrast, an exploratory observation we do not claim here: with all tags, Llama kept the request act for 1/10 plain politeness targets against 10/10 polite ones. [PENDING: W-B transfer report exists (interim wb1i): add random and enriched AACText estimates separately, defaults only on confirmed-ambiguous inputs, violations only on supplied commitments, with audit uncertainty.] Until then, the only corpus evidence is that a four-word content compressor

TABLE II

SELECTED COUNTEREXAMPLES, ALL STORED OUTPUTS (STRINGS RE-READ FROM THE RUN FILES BY `MAKE_MACROS.PY`; IDS TRUNCATED). SELECTION RULES (P0 UNLESS STATED; FAMILIES BY ID, MODELS IN THE ORDER QWEN, LLAMA, GEMMA, TERRA): E1 FIRST NEGATED POLARITY TARGET, ITS BARE/NONCRIT/CHIP1 OUTPUTS; E3 FIRST PLAIN POLITENESS TARGET MADE POLITE UNDER FULL INPUT; E4 FIRST QUESTION TARGET MADE A REQUEST UNDER CHIP1, QUARANTINE EXCLUDED; E5/E6 FIRST FAILURE→PASS AND FAILURE→FAILURE GUARD PAIRS BY THE INTERVENTION REPORT'S HASH ORDER, QUARANTINE EXCLUDED; E10 FIRST RAW-ORACLE FAILURE ON A NORMALIZED SOURCE REPRODUCTION (QUARANTINED FAMILY, ORACLE DIAGNOSTIC ONLY). THE RIGHT COLUMN GIVES THE RATE IN EACH RULE'S ELIGIBLE STRATUM; A SELECTED FAILURE IS NOT TYPICAL. IN *typed* ROWS THE SOURCE IS VISIBLE TO THE GENERATOR; EXTRACTION IS ALWAYS BLIND.

Example	Visible input	Stored output	Model, family, prompt, channel; ID	Eligible-stratum rate
E3 full input	*typed:* `Wear headphones for your music.`	"Please wear headphones for your music."	llama, pol2-07/B, P0, full; `30caad83`	plain→polite 9/40 (pooled); this model 1/10; reverse 2/40
E1 bare	`like medication`	"I like medication"	qwen, pol-07/A, P0, bare; `bba84cad`	affirmative in 40/40 bare outputs (default, not a failure)
E1 noncrit	`like new medication`	"I like new medication"	qwen, pol-07/A, P0, noncrit; `ce2e2b1e`	length control / default-sensitivity diagnostic
E1 chip1	`[NOT] like medication`	"I don't like medication."	qwen, pol-07/A, P0, chip1; `e6f09261`	negated cue kept 40/40
E4 chip1	`[?] bring dog`	"Will you bring the dog?"	llama, act-07/A, P0, chip1; `4cf4eb25`	question kept 22/28; became request 3/28
E5 guard	`[PLAIN] let finish sentence`	"Please let me finish my sentence." → "Let me finish the sentence."	gemma, pol2-14/A, P0→G1, chip1; `7aaf1cd7`	repaired: 63/80 cue-channel failures
E6 guard	`[?] [YOU] charge wheelchair`	"Can you charge my wheelchair?" → "Can you charge my wheelchair?"	llama, act-11/A, P0→G1, chipall; `5c224e70`	still failing: 17/80
E10 oracle	*typed:* `Yes, you can borrow my car this weekend.`	"Yes, you can borrow my car this weekend."	qwen, act-16/B, P0, full; `b6b079ac`	raw oracle failed 14/2,292 normalized identity outputs; quarantined

TABLE III

REQUIREMENT → EXECUTABLE CHECK → DEVELOPER DECISION → FAILURE ENVELOPE. PRIMARY POOL (58 FAMILIES), MACHINE-SCORED; GUARD RATES ARE CUE-CHANNEL TARGETS.

Requirement	Check and finding	Developer decision (evaluated)	Failure envelope
R2: a typed sentence keeps its scored values	MR3: P0 contract pass 449/464; plainness lost in 9/40 directives	Route an explicitly selected full mode to exact copy: 116/116 source identities pass, zero model calls. Guard alternative: 463/464	Copy applies only when the user declares full mode; it does not detect whether keywords already form a sentence, and expands nothing.
R1: a supplied cue is honored	MR2: 440/464 chip1 match; cue-channel strict pass (incl. no added politeness) 94.3%	Runtime guard: rule-based check against supplied cues, one feedback regeneration on a flag. Repairs 63/80 [68.4, 88.9]% failures; strict pass 94.3%→98.8%; 0.104 extra calls and 0.17 s per input in our serving setup. Strip-only baseline repairs 34/80	17 failures remain, mostly speech act (E6); 14/1,312 correct outputs changed, 0 became failures; repairs are scored predicates, not fluent speech ("Ask you to wear headphones." passes); monitor and evaluator share layer0.
Oracle: an unchanged source passes (MR0)	Identity on stored outputs: 14 false fails, one sentence	Relax acceptance; re-test with keyed mutants; on failure (4/6), quarantine the two families	Refusal/acceptance unresolved; idiomatic negation and vague quantity are shared machine blind spots; human gate pending.
R3: defaults are reported	Bare: 40/40 affirmative; P2 covers both in 10/40	Document the default and price the other reading; expose a negation control	Nominal selections only: 21/284 misses not repaired by the tag; no user effort measured.

dropped at least one explicit cue, by a rule-based detector, in 59.2% of 6,142 crowd-written fictional AAC messages [22], ignoring pronoun loss.

VII. FROM REQUIREMENT TO DEVELOPER DECISION

Table III shows how each requirement becomes a check, what the check found, what a developer could decide, and what the evidence does not cover.

The full-mode row is the clearest engineering lesson. If an interface lets the person say "this is my sentence", the requirement is met by not generating at all; the remaining question is a product one, whether full mode is offered and discoverable. The guard row shows that the same requirements can run at run time, with measured repairs, residual failures and cost. The oracle row shows why the checks need their own regression tests: our first repair silently weakened them.

Recommendations: Write each requirement conditional on the input and per dimension; "preserve meaning" is not testable, "if the input carries `[NOT]`, the output is negated" is. Test with minimal pairs, report joint as well as focal preservation, and keep defaults separate from failures. Treat added politeness or certainty on full input as a failing test and make style assistance opt-in. Give the oracle identity and keyed-mutation tests, and re-run them after every relaxation. Compare labels, not strings, and repeat draws for any release gate.

VIII. Threats to Validity

Construct. Labels are coarse, text-explicit categories from a codebook; "plain" means no explicit marker, and coverage of content, referents, negation scope and certainty degree (emphatic outputs pass a plain target by design) is partial. *Oracle.* All rates depend on an LLM extractor that has not yet passed the human gate; machine-panel agreement shares errors. *Selection and scale.* Ten authored families per dimension and four fixed models support counterexamples and regression tests, not prevalence; examples were selected after discovery, by declared rules. The politeness asymmetry is prompt-dependent (P1) and model-dependent. *Follow-ups.* Composition evidence uses eight scenarios and a tag that models literalize. *Cost.* Selection counts omit reading, navigation and fatigue [23]. *Serving.* Guard latency used a local server with partial CPU offload and is not a deployment estimate. *No users.* We cannot speak to benefit, agency or whether anyone wants a given output; we do not assume an "autistic style" of language [24], [25].

IX. Conclusion

Even a sentence the person typed in full can come back changed. Conditioning requirements on what the input carried separates that violation from the guesses an interface forces when it collects less, and turns both into executable checks with a developer decision attached. Testing the oracle mattered as much as testing the models: a repair that removed false alarms also removed fault sensitivity. We release the families, requirements, harness, oracle tests, stored outputs and analysis code.[PENDING: repository URL / release snapshot]

Ethics

No data were collected from AAC users; AACText is public under CC BY 4.0. Human annotators label model outputs under a fixed codebook.[PENDING: institutional determination; annotator compensation; adviser participation only as actually completed]

Acknowledgment

[PENDING: IEEE AI disclosure: name the AI system(s) that drafted text, figures and code, the sections affected and the level of use (VENUE.md). The models under test, the extractor and the arbiter are research instruments described in Sections IV–V.]

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